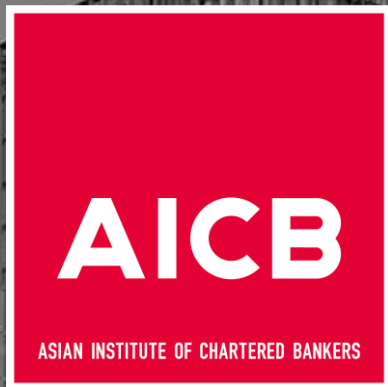




AI GOVERNANCE FRAMEWORK - An initiative by the Chief Risk Officers' Forum (CRO's)

■ Objectives of the Framework

Objectives :

To establish an industry-led guideline* and best practices in managing AI risks, incorporating:

- **Balanced perspectives** managing risk and innovation
- **Practical considerations** for operationalisation
- **Consensus building** with stakeholders across the financial services industry

Industry-driven framework that aims at ::

1. Setting **broad principles** for AI Adoption
2. Minimizing **potential risks** related to AI adoption
3. Supporting **AI innovation** in financial services
4. **Addressing challenges** in AI application
5. Aligning AI principles **internationally**

Note:

** - The AI Governance Framework serves as guidance only, and its adoption by banks is at their discretion.*



SCOPE AND DEFINITION OF AI

■ Scope and Definition of AI

CROs Forum proposes to adopt the following definition elements; i.e: what constitutes AI

- + Simulation of human intelligence by machines
- + Able to learn and acquire knowledge
- + Has adaptive elements

Consensus suggest also to avoid listing exclusions to specific algorithms or models

- Criteria for an AI system should be evaluated based on 3 conditions listed on the left **without rigid or explicit exclusions**

Scope of AI governance

- Scope should focus on **predictive and probabilistic models (for all AI models, not just risk models)** over simple rule-based systems without learning, adaptive components



GUIDING PRINCIPLES

Summary of Agreed Guiding Principles

1. Fairness	2. Ethical	3. Accountability	4. Transparency	5. Explainability	6. Reliability	7. Security
- AI-driven decisions do not discriminate or create harmful bias against any individual or group of individuals Differential treatments between different individuals or groups of people should be justifiable - The use of personal attributes as input factors for AI-driven decisions are justifiable	- Use of AI complies with all relevant laws and regulations - Use of AI is aligned to the financial institution's corporate values, ethical standards, and code of conduct	- Board and senior management are accountable for all AI strategies and outcomes - Use of AI for decision-making is approved by an appropriate, competent internal authority - Financial institutions are accountable for both internally developed and externally sourced AI applications	Use of AI is sufficiently disclosed and communicated to those affected where appropriate, commensurate to level of materiality, without revealing proprietary or security-critical information	- There should be sufficient level of explainability over the underlying design and intent of AI applications, commensurate to the level of materiality, without revealing proprietary or security-critical information - AI system processes should be well-documented and traceable, with clear and understandable records of decisions, datasets and processes used to yield the system's decisions. - When explainability is limited or difficult, it should be supported by valid justifications	Trained AI models are adequately tested and validated to ensure accuracy, relevance, and robustness in handling errors, invalid inputs, or stressful conditions for deployment and ongoing use	- Maintain privacy, data protection, and security throughout the data lifecycle, employing robust data governance and management - Implement appropriate controls and security measures to ensure the robust security and resilience of AI systems and to protect against new threats - Comply with relevant laws and regulations, and implement appropriate controls where possible to protect data

1. Organisations may consider incorporating these principles into their own corporate principles, where relevant and desired

■ Principle 1: Fairness

Principle statements	Key considerations
1. AI-driven decisions do not discriminate or create harmful bias against any group of individuals 2. Differential treatments between different individuals or groups of people should be justifiable 3. The use of personal attributes as input factors for AI-driven decisions are justifiable	<u>Good practices to ensure fairness</u> • Ensure alignment to BNM Fair Treatment of Financial Consumer guidelines • Ensure representativeness of input data used • Establish fairness assessment metrics during testing, validation and reassessment of AI models to detect and address bias • Avoid incorporating personal attributes as factors in AI models if their use raises ethical or moral concerns, even when they are predictive • Develop and include monitoring and accounting mechanisms to avoid unintentional discrimination when implementing decision-making systems • Promote interdisciplinary collaboration among diverse stakeholders throughout AI lifecycle to ensure adequate representation • Establish clear guidelines on the use of personal attributes taking into consideration relevance for use, legal mandates for use, and guidelines to minimise potential harms <u>Example of fairness principle applied:</u> • <u>Chatbot</u>: Interactions provide consistent answers to standard inquiries regardless of user's demographics, and collects only essential personal data for service execution e.g. requesting last-4 digits of IC for account verification • <u>AI credit scoring</u>: Credit assessment based on objective financial data e.g. debt-to-income ratio, not subjective personal characteristics like race or gender

■ Principle 2: Ethical

Principle statements	Key considerations
1. Use of AI complies with all applicable laws and regulations 2. Use of AI is aligned to the bank's corporate values, ethical standards, and code of conduct	<u>Good practices to ensure ethical use</u> • Ensure relevant stakeholders are included and consulted in the AI development process to ensure adequate consideration given for ethical practices • Ensure relevant forum assigned with AI governance and risk oversight comprises of diverse stakeholders of relevant expertise and independent representatives so as to ensure review of ethical use of AI systems <u>Example of ethical principle applied:</u> • <u>Chatbot</u>: Complies with data privacy i.e. no unnecessary data collection and interactions do not provide responses that are morally questionable or break the law • <u>AI credit scoring</u>: Complies with all BNM financial regulations on credit lending and no use of criteria that are morally questionable or goes against the bank's corporate values e.g. race

■ Principle 3: Accountability

Principle statements	Key considerations
1. Board and senior management are accountable for all AI strategies and outcomes 2. Use of AI for decision-making is approved by an appropriate, competent and qualified internal authority 3. Financial institutions are accountable for both internally developed and externally sourced AI applications	<u>Good practices to ensure accountability</u> • Establish robust framework to govern and enforce accountability, including designing and assigning responsibilities appropriately at each stage across the AI lifecycle • Accountability over externally sourced AI applications should be aligned with BNM RMiT guidelines on third party service provider management and BNM Outsourcing guidelines <u>Example of accountability principle applied:</u> • <u>Chatbot</u>: Management approved use of external AI chatbot after careful due diligence, limiting to basic enquiries, with complex matters escalated to human support, and establishing SLAs for regular oversight to monitor chatbot's performance over time using customer surveys • <u>AI credit scoring</u>: Implementation is approved by Board and Management risk committees, ensuring AI is only used to augment, not replace, human decision-making for credit approvals

■ Principle 4: Transparency

Principle statements	Key considerations
1. Use of AI is sufficiently disclosed and communicated to those affected where appropriate, commensurate to level of materiality, without revealing proprietary or security-critical information	<u>Good practices to ensure transparency</u> • Transparency should be appropriately balanced to enhance public trust while avoiding confusion, IP exposure or opportunities for exploitation of proprietary code • Upon request, disclosures should help individuals understand what data is used in AI-driven decisions about them and how the data affects the decision • Level of transparency should consider the materiality of the AI application and impact on data subjects <u>Example of transparency principle applied:</u> • <u>Chatbot</u>: Use and capabilities are disclosed and communicated clearly in statement of use before user begins interaction • <u>AI credit scoring</u>: Upon request, use of AI is sufficiently disclosed, and how it assisted the human decision-making process in approving or rejecting a loan application

■ Principle 5: Explainability

Principle statements	Key considerations
1. There should be sufficient level of explainability over the underlying design and intent of AI applications, commensurate to the level of materiality, without revealing proprietary or security-critical information 2. AI system processes should be well-documented and traceable, with clear and understandable records of decisions, datasets and processes used to yield the system's decisions. 3. When explainability is limited or difficult, it should be supported by valid justifications For when technical explainability is limited: • Impose more testing or pilot in a control environment before releasing in full production • Describing how predictions play a role in the decision-making process • Describe using implicit explanations of how the algorithm functions instead of explicit description of the model's logic • Demonstrate repeatability to provide larger degree of confidence of AI system's consistency • Disclose cases where explainability may not be practical e.g. AML/fraud detection patterns or proprietary information	<u>Good practices to improve explainability</u> i. Document how the model training and selection processes are conducted, the reasons for how the decisions are made and measures taken to address identified risks. ii. Incorporate descriptions of the solutions' design and system technical specifications into documentation. iii. If AI system was procured from a third-party, to request assistance from the third-party AI service provider to explain how the solution functions iv. Using supplementary explanation tools to help clarify less interpretable 'black-box' systems, by making the rationale behind their outputs more understandable to users <u>Example of explainability principle applied:</u> • <u>Chatbot</u>: Customer service team fully informed on chatbot query processing and response mechanisms, ensuring alignment with bank protocols • <u>AI credit scoring</u>: Credit scoring decisions are documented including data and algorithms used for continuous improvement and audit purposes

■ Principle 6: Reliability

Principle statements	Key considerations
1. Trained AI models are adequately tested and validated to ensure accuracy, relevance, and robustness in handling errors, invalid inputs, or stressful conditions for deployment and ongoing use	<u>Good practices to ensure reliability</u> • Implement rigorous testing, validation and reassessment of AI models, including stress tests with extreme scenarios, to understand system behaviour and ensure robustness under various conditions • Conduct repeatability assessments in live environments and assess how exceptions can be identified and handled when decisions are not repeatable • Perform continuous monitoring and re-assessment of AI models, especially those trained on time-sensitive data to ensure ongoing relevance and technical performance • Implement human oversight and final determination for AI systems making high-risk and material-impact decisions <u>Example of reliability principle applied:</u> • <u>Chatbot</u>: Chatbot stress-tested against variety of user inquiries including extreme scenarios to ensure responses comply with the bank's customer service approved protocols • <u>AI credit scoring</u>: Credit risk assessment stress tested against different economic conditions and factors to ensure consistency

■ Principle 7: Security

Principle statements	Key considerations
1. Maintain privacy, data protection, and security throughout the data lifecycle, employing robust data governance and management 2. Implement appropriate controls and security measures to ensure the robust security and resilience of AI systems and to protect against new threats 3. Comply with relevant laws and regulations, and implement appropriate controls where possible to protect data	<u>Good practices to ensure security</u> • Establish controls and contingency measures, including AI inventory and risk mapping, incident management, change management, and response plans, application deactivation, and fallback procedures, proportionate to the materiality of risks and specific use cases, to address failures, deviations from intended use, or breaches in AI usage • Define the period of the data used in LLM training to ensure relevance and accuracy of the output, but it also presents heightened risks in the event of a breach • Implement privacy and security controls, as well as related safeguards in AI, including open-source tools, to address threats such as unauthorised usage, data or model manipulation, data breaches and malicious attacks • Understand the risks of adopting AI and considerations around deployment models including third-party managed platforms, internally built on-premises platforms, or internally built cloud-based platforms, based on the associated risks • Evaluation of data-cominggling between different entities to maintain data integrity and privacy • Establish documentation of datasets, processes, decisions and outcomes throughout the AI system lifecycle • To consider the implementation of data protection controls including data anonymization, tokenisation, encryption, and access controls, ensuring that personal data is safeguarded throughout the AI lifecycle • To consider the requirement related data retention, data lifecycle management, and historical data in relation to AI deployment



GOVERNANCE AND ACCOUNTABILITY

■ Governance and Accountability

Proposal

- + Establishing **clear accountability rules and oversight requirements** for Board and Senior Management
- + **Avoiding prescribing** dedicated roles, committees and reporting lines – keep flexibility for organisation to define structure
- + Units or individuals appointed should **possess suitable qualifications and appropriate competence** for their appointed roles
- + Organisations can choose to **leverage on existing** internal governance structures and/or **implement new structures** if necessary

■ Roles and Responsibilities of the Board

- + Holds **oversight accountability for AI-driven outcomes** made by the financial institution
- + Ensure and oversee the adequacy of the financial institution's **AI strategic plans**
- + Ensure a proper **AI governance and risk management framework** is established to oversee the responsible, ethical use and effective management of AI applications throughout their life cycle within the financial institution
- + Ensure **independent validation** is conducted to verify the effectiveness and compliance of AI applications with established AI governance and risk management framework
- + Allocate **sufficient time to discuss** AI-risk related issues and emerging threats.
- + Designate a **board-level committee** (new or existing) to oversee material matters related to AI governance and risk

■ Roles and Responsibilities of Senior Management

- + Accountable for **AI-driven outcomes** made by the financial institution
- + Integrate board-approved AI governance and risk management framework requirements into **policies and procedures** to ensure effective governance and management of AI risks
- + Ensure the **adequate allocation of resources** to maintain robust AI implementations and appropriately skilled and competent staff to support the effective management of AI risk and ethical use of AI
- + Define and assign **clear responsibilities** for the validation, testing and ongoing monitoring of AI applications to ensure their accuracy, reliability and appropriateness for operational use
- + Establish appropriate oversight arrangements to monitor the **performance and compliance** of AI applications, ensuring they adhere to regulatory requirements
- + Ensure **training and awareness programmes** are implemented to promote Board and staff awareness of AI technologies, associated risks, and their roles in managing these risks
- + **Regularly report to the board** on AI-related activities, including risk management, operational effectiveness, and any incidents or challenges encountered



IMPLEMENTATION CONSIDERATIONS

■ Implementation Considerations (1/2)

1. Assign **clear responsibilities** for the oversight and management of AI applications across different levels and functions, with appropriate governance and reporting arrangements
2. Ensure **sufficient investment and resources** are allocated to build necessary capabilities for managing AI risks responsibly and ensuring ethical AI practices
3. Implement appropriate **training and awareness programmes** to ensure staff have adequate knowledge, skills and awareness to maintain the continued usage of AI and/or to work in an environment using AI
4. Adopt a **risk-based approach to integrating human oversight** into AI with due consideration given to evaluate the risk level and potential impacts of the application involved
5. Establish **comprehensive risk management processes and controls** to assess, manage and mitigate risks throughout the lifecycle of AI.
6. Conduct **testing, evaluation, verification and validation of AI models** to confirm accuracy, relevance and technical robustness against errors, invalid inputs or stressful conditions before deployment for production use.
7. Perform **periodic reviews and continuous monitoring** of AI evaluating risk metrics to ensure they remain effective, reliable, and true to their intended purpose, adjusting for any shifts in model behavior or performance over time.

Implementation Considerations (2/2)

8. Establish **controls and contingency measures**, including AI inventory and risk mapping, incident management, change management, and response plans, application deactivation, and fallback procedures, proportionate to the materiality of risks and specific use cases, to address failures, deviations from intended use, or breaches in AI usage
9. Oversee and implement measures that enforces **effective execution of oversight** where AI systems or services are procured from and/or managed by external parties.
10. Implement **privacy and security controls, as well as related safeguards** in AI, including open-source tools, to address threats such as unauthorised usage, data or model manipulation, data breaches and malicious attacks
11. Define the **period of the data used** in LLM training to ensure relevance and accuracy of the output, but it also presents heightened risks in the event of a breach
12. Understand the **risks of adopting AI and considerations around deployment models** including third-party managed platforms, internally built on-premises platforms, or internally built cloud-based platforms, based on the associated risks
13. Evaluation of **data commingling between different entities** to maintain data integrity and privacy
14. Establish **documentation of datasets, processes, decisions and outcomes** throughout the AI system lifecycle
15. To consider the **implementation of data protection controls** including data anonymization, tokenisation, encryption, and access controls, ensuring that personal data is safeguarded throughout the AI lifecycle
16. To consider the requirement related **data retention, data lifecycle management, and historical data** in relation to AI deployment



Thank You